

ENTERPRISE ACTIVE DIRECTORY DEPLOYMENT AND SECURITY AUDIT

Security Lab | Windows Server 2022 | GPO Hardening | Kali Linux Assessment

```
C:\Windows\system32\cmd.exe
PS C:\Users\Administrator> Get-ADDomain | Select-Object Name, DomainMode, Forest, InfrastructureMaster

Name          DomainMode Forest      InfrastructureMaster
-----
abc Windows2016Domain abc.local DC-01.abc.local

PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName

Name          DistinguishedName
-----
Domain Controllers OU=Domain Controllers,DC=abc,DC=local
ABC-Technologies OU=ABC-Technologies,DC=abc,DC=local
Security      OU=Security,OU=ABC-Technologies,DC=abc,DC=local
IT            OU=IT,OU=ABC-Technologies,DC=abc,DC=local

PS C:\Users\Administrator> Get-ADUser -Filter * -Properties Department, MemberOf | Select-Object Name, SamAccountName, Department, Enabled

Name          SamAccountName Department      Enabled
-----
Administrator Administrator True
Guest         Guest          False
krbtgt        krbtgt         False
Aayush        aayush        IT-Security    True
Ankit         ankit         Cloud Operations True
Abhinav       abhinav       Network-Security True
Ahtesham      ahtesham      Monitoring     True
```

Windows Server 2022 & AD DS Setup

Domain Provisioning, OU Structure and Core Services Verification

POWERSHELL

Domain, Organisational Units & USer Accounts

```
PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName

Name                DistinguishedName
----                -
Domain Controllers  OU=Domain Controllers,DC=abc,DC=local
ABC-Technologies    OU=ABC-Technologies,DC=abc,DC=local
Security            OU=Security,OU=ABC-Technologies,DC=abc,DC=local
IT                  OU=IT,OU=ABC-Technologies,DC=abc,DC=local

PS C:\Users\Administrator> Get-ADUser -Filter * -Properties Department, MemberOf | Select-Object Name, SamAccountName, Department, Enabled

Name                SamAccountName Department Enabled
----                -
Administrator Administrator      True
Guest               Guest                False
krbtgt              krbtgt               False
Aayush              aayush              IT-Security   True
Ankit                ankit                Cloud Operations True
Abhinav              abhinav              Network-Security True
Ahtesham             ahtesham             Monitoring    True

PS C:\Users\Administrator> Get-GPO -All | Select-Object DisplayName, GPOStatus, CreationTime, ModificationTime

DisplayName                GpoStatus CreationTime      ModificationTime
-----
Default Domain Policy      AllSettingsEnabled 7/15/2026 11:40:06 AM 7/23/2026 11:54:08 PM
Default Domain Controllers Policy AllSettingsEnabled 7/15/2026 11:40:06 AM 7/15/2026 11:40:06 AM
GPO_Restrict_ControlPanel AllSettingsEnabled 7/23/2026 11:59:17 PM 7/23/2026 11:59:16 PM
Map Network Drive 2        AllSettingsEnabled 7/19/2026 12:35:20 PM 7/21/2026 12:16:40 PM
GPO_Disable_CMD            AllSettingsEnabled 7/26/2026 8:57:08 AM 7/26/2026 8:57:08 AM

PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * | Get-GPInheritance | Select-Object Path, GpoLinks

Path                GpoLinks
----                -
ou=domain controllers,dc=abc,dc=local {Default Domain Controllers Policy}
ou=abc-technologies,dc=abc,dc=local {}
ou=security,ou=abc-technologies,dc=abc,dc=local {}
ou=it,ou=abc-technologies,dc=abc,dc=local {}
```

POWERSHELL

Static IP, DNS Configuration & Core Services

```
Default Domain Controllers Policy AllSettingsEnabled 7/15/2026 11:40:06 AM 7/15/2026 11:40:06 AM
GPO_Restrict_ControlPanel        AllSettingsEnabled 7/23/2026 11:59:17 PM 7/23/2026 11:59:16 PM
Map Network Drive 2              AllSettingsEnabled 7/19/2026 12:35:20 PM 7/21/2026 12:16:40 PM
GPO_Disable_CMD                  AllSettingsEnabled 7/26/2026 8:57:08 AM 7/26/2026 8:57:08 AM

PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * | Get-GPInheritance | Select-Object Path, GpoLinks

Path                GpoLinks
----                -
ou=domain controllers,dc=abc,dc=local {Default Domain Controllers Policy}
ou=abc-technologies,dc=abc,dc=local {}
ou=security,ou=abc-technologies,dc=abc,dc=local {}
ou=it,ou=abc-technologies,dc=abc,dc=local {}

PS C:\Users\Administrator> Get-NetIPAddress -AddressFamily IPV4 | Where-Object {$_.InterfaceAlias -notlike "*Loopback*"} | Select-Object InterfaceAlias, IPAddress, PrefixLength

InterfaceAlias IPAddress      PrefixLength
-----
Ethernet0      192.168.10.30 24
Ethernet0      192.168.10.10 24

PS C:\Users\Administrator> Get-DnsClientServerAddress -AddressFamily IPV4 | Select-Object InterfaceAlias, ServerAddresses

InterfaceAlias      ServerAddresses
-----
Ethernet0            {127.0.0.1}
Loopback Pseudo-Interface 1 {}

PS C:\Users\Administrator> Get-Service -Name NTDS, DNS, KDC, LanmanServer | Select-Object Name, DisplayName, Status

Name                DisplayName                Status
----                -
DNS                  DNS Server                 Running
KDC                  Kerberos Key Distribution Center Running
LanmanServer         Server                     Running
NTDS                 Active Directory Domain Services Running
```

CORE SETUP

Deployed **DC-01.abc.local** with dedicated departmental OUs to support Role-Based Access Control (RBAC)

User Provisioning & Security Group Mapping

Implementing Role-Based Access Control (RBAC) across Departments

Aayush --- IT-Security

```
C:\Users\Aayush>whoami & systeminfo | findstr /B /C:"Domain" & gpreresult /r
abc\Aayush
Domain:                abc.local

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Created on   8/ 1/ 2026 at 10:34:00 AM

-----
RSOP data for ABC\Aayush on DESKTOP-BQSH4MS : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.26200
Site Name:              N/A
Roaming Profile:        N/A
Local Profile:          C:\Users\Aayush
Connected over a slow link?: No

USER SETTINGS
-----
CN=Aayush,CN=Users,DC=abc,DC=local
Last time Group Policy was applied: 8/1/2026 at 10:30:28 AM
Group Policy was applied from:      DC-01.abc.local
Group Policy slow link threshold:    500 kbps
Domain Name:                        ABC
Domain Type:                        Windows 2008 or later

Applied Group Policy Objects
-----
Map Network Drive 2
GPO_Restrict_ControlPanel
GPO_Disable_CMD
```

Abhinav --- Network Security

```
C:\Users\Abhinav>whoami & systeminfo | findstr /B /C:"Domain" & gpreresult /r
abc\Abhinav
Domain:                abc.local

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-----
RSOP data for ABC\Abhinav on DESKTOP-BQSH4MS : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.26200
Site Name:              N/A
Roaming Profile:        N/A
Local Profile:          C:\Users\Abhinav
Connected over a slow link?: No

USER SETTINGS
-----
CN=Abhinav,CN=Users,DC=abc,DC=local
Last time Group Policy was applied: 8/1/2026 at 10:20:45 AM
Group Policy was applied from:      DC-01.abc.local
Group Policy slow link threshold:    500 kbps
Domain Name:                        ABC
Domain Type:                        Windows 2008 or later

Applied Group Policy Objects
-----
Map Network Drive 2
GPO_Restrict_ControlPanel
GPO_Disable_CMD
```

Ankit --- Cloud Security

```
C:\Users\Ankit>whoami & systeminfo | findstr /B /C:"Domain" & gpreresult /r
abc\Ankit
Domain:                abc.local

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Created on   8/ 1/ 2026 at 10:28:32 AM

-----
RSOP data for ABC\Ankit on DESKTOP-BQSH4MS : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.26200
Site Name:              N/A
Roaming Profile:        N/A
Local Profile:          C:\Users\Ankit
Connected over a slow link?: No

USER SETTINGS
-----
CN=Ankit,CN=Users,DC=abc,DC=local
Last time Group Policy was applied: 8/1/2026 at 10:27:00 AM
Group Policy was applied from:      DC-01.abc.local
Group Policy slow link threshold:    500 kbps
Domain Name:                        ABC
Domain Type:                        Windows 2008 or later
```

Ahtesham --- Monitoring

```
C:\Users\Ahtesham>whoami & systeminfo | findstr /B /C:"Domain" & gpreresult /r
abc\Ahtesham
Domain:                abc.local

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Created on   8/ 1/ 2026 at 10:03:34 AM

-----
RSOP data for ABC\Ahtesham on DESKTOP-BQSH4MS : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.26200
Site Name:              N/A
Roaming Profile:        N/A
Local Profile:          C:\Users\Ahtesham
Connected over a slow link?: No

USER SETTINGS
-----
CN=Ahtesham,CN=Users,DC=abc,DC=local
Last time Group Policy was applied: 8/1/2026 at 10:01:01 AM
Group Policy was applied from:      DC-01.abc.local
Group Policy slow link threshold:    500 kbps
Domain Name:                        ABC
Domain Type:                        Windows 2008 or later

Applied Group Policy Objects
-----
Map Network Drive 2
GPO_Restrict_ControlPanel
GPO_Disable_CMD
```

Group Policy Objects & Workstation Hardening

Enforcing Security Policies and Workstation Restraints

GPMC

Group Policy Object Inventory & OU Inheritance

```
PS C:\Users\Administrator> Get-GPO -All | Select-Object DisplayName, GPOStatus, CreationTime, ModificationTime

DisplayName                                GpoStatus CreationTime      ModificationTime
-----
Default Domain Policy                    AllSettingsEnabled 7/15/2026 11:40:06 AM 7/23/2026 11:54:08 PM
Default Domain Controllers Policy        AllSettingsEnabled 7/15/2026 11:40:06 AM 7/15/2026 11:40:06 AM
GPO_Restrict_ControlPanel                AllSettingsEnabled 7/23/2026 11:59:17 PM 7/23/2026 11:59:16 PM
Map Network Drive 2                      AllSettingsEnabled 7/19/2026 12:35:20 PM 7/21/2026 12:16:40 PM
GPO_Disable_CMD                          AllSettingsEnabled 7/26/2026 8:57:08 AM 7/26/2026 8:57:08 AM

PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * | Get-GPInheritance | Select-Object Path, GpoLinks

Path                                GpoLinks
-----
ou=domain controllers,dc=abc,dc=local {Default Domain Controllers Policy}
ou=abc-technologies,dc=abc,dc=local {}
ou=security,ou=abc-technologies,dc=abc,dc=local {}
ou=it,ou=abc-technologies,dc=abc,dc=local {}

PS C:\Users\Administrator> Get-NetIPAddress -AddressFamily IPV4 | Where-Object {$_.InterfaceAlias -notlike "**Loopback**"} | Select-Object InterfaceAlias, IPAddress, PrefixLength

InterfaceAlias  IPAddress      PrefixLength
-----
Ethernet0       192.168.10.30  24
Ethernet0       192.168.10.10  24

PS C:\Users\Administrator> Get-DnsClientServerAddress -AddressFamily IPV4 | Select-Object InterfaceAlias, ServerAddresses

InterfaceAlias  ServerAddresses
-----
Ethernet0       {127.0.0.1}
Loopback Pseudo-Interface 1 {}
```

WINDOWS 11

Client RSOP & Policy Inheritance

```
USER SETTINGS
-----
CN=Ahtesham,CN=Users,DC=abc,DC=local
Last time Group Policy was applied: 8/1/2026 at 10:01:01 AM
Group Policy was applied from:      DC=01.abc.local
Group Policy slow link threshold:   500 kbps
Domain Name:                        ABC
Domain Type:                        Windows 2008 or later

Applied Group Policy Objects
-----
Map Network Drive 2
GPO_Restrict_ControlPanel
GPO_Disable_CMD

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----
Domain Users
Everyone
BUILTIN\Users
NT AUTHORITY\INTERACTIVE
CONSOLE LOGON
NT AUTHORITY\Authenticated Users
This Organization
LOCAL
Authentication authority asserted identity
Medium Mandatory Level
```

5 GPOS actively enforced Domain-Wide

4 OUs mapped to departmental policy scope

100% Policy inheritance confirmed on client RSOP

HARDENING STATUS

Successfully pushed centralised administrative constarints to end-user worksatations via GPOs

PHASE 04

Kali Linux Security Audit & Enumeration

NMAP Service & Port Scan – 192.168.10.30

7 Domain accounts enumerated via authenticated SMB

Service enumeration confirmed exposed AD ports and validated user account discoverability

Authenticated SMB User Enumeration

```
(root@HACTIVIST2009)-[/home/bjorn]
# netexec smb 192.168.10.30 -u 'abhinav' -p 'emp123!' --users
SMB 192.168.10.30 445 DC-01 [*] Windows Server 2022 Build 20348 x64 (name:DC-01) (domain:abc.local) (signing:True) (SMBv1:None) (Null
Auth:True)
SMB 192.168.10.30 445 DC-01 [+] abc.local\abhinav:emp123!
SMB 192.168.10.30 445 DC-01 -Username- -Last PW Set- -BadPW- -Description-
SMB 192.168.10.30 445 DC-01 Administrator 2026-07-14 18:15:50 0 Built-in account for administering the computer/
domain
SMB 192.168.10.30 445 DC-01 Guest <never> 0 Built-in account for guest access to the compute
:/domain
SMB 192.168.10.30 445 DC-01 krbtgt 2026-07-15 18:41:28 0 Key Distribution Center Service Account
SMB 192.168.10.30 445 DC-01 aayush 2026-07-19 06:51:01 0
SMB 192.168.10.30 445 DC-01 ankit 2026-07-19 07:06:15 0
SMB 192.168.10.30 445 DC-01 abhinav 2026-07-19 07:09:25 0
SMB 192.168.10.30 445 DC-01 ahtesham 2026-07-19 07:10:48 0
SMB 192.168.10.30 445 DC-01 [*] Enumerated 7 local users: ABC
```